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Designing Resilient Factory Layouts Through Data-Driven Scenario Analysis with an Industrial Application

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Abstract. Developing resilient and robust facility layouts is a long-term strategic problem for contemporary manufacturing systems, particularly in settings with operational uncertainty and spatial complexity. When departments are assigned to multiple layout blocks of different area requirements and material flows are prone to disruption, conventional deterministic models are not able to provide robust solutions. This paper bridges this gap by introducing a scenario-based robust facility layout model as a mixed-integer linear programming (MILP) problem. We introduce an MILP formulation that couples the unequal-area facility layout problem with a capacitated network flow problem. Unlike standard models that assume direct paths, this approach explicitly models interblock logistics through capacity-constrained gateways (ramps). The model employs a penalty-based robust objective function that integrates expected cost minimization with stability deviations and soft capacity constraints, allowing the identification of feasible solutions even under compound disruption scenarios where rigid models would fail. The proposed approach is demonstrated on a real-life application case from a leading European home appliance manufacturer based in Turkey, where the plant layout must fit changing operational conditions and capacity restrictions. Leveraging actual production data and expert knowledge such as observations of material surges and production shutdowns, realistic disruption scenarios are defined for the model to develop solutions that remain feasible and efficient in different operating states. Simulation-based verification shows that optimized layouts preserve robust performance under dynamic and uncertain flow conditions. The proposed layout reduced total material handling distance by 19.3% and increased production area utilization by 31% compared with the current layout. The study provides a scalable and data-driven approach to industrial facility planning and demonstrates the practical applicability of robust optimization to increase resilience under uncertainty in production settings.

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Introduction

The facility layout problem (FLP) is a classical area of research in industrial engineering and operations research that has a direct influence on the performance of manufacturing systems. FLP is concerned with finding the most appropriate location for departments, machines, or workstations in a facility to reduce material handling costs and improve overall production performance (Deshpande et al. 2016, Sun et al. 2018). An effective layout eliminates unnecessary movement, improves production capacity, minimizes energy consumption and resource utilization, and ultimately reduces operational costs (Ku et al. 2011, Navidi et al. 2012). In contrast, ineffective layouts may result in higher transportation costs and lower production capacity, eventually increasing production costs. Even a small improvement in layout design can bring about quantifiable gains.

Previously, the majority of FLP research made simplifying assumptions, for example, equally sized departments or deterministic flow matrices. More recent work shows that such assumptions do not reflect the complexity of operational processes in contemporary manufacturing settings. In practice, departments are different in shape and size and have to meet a variety of operational constraints (Loiola et al. 2007, Grobelny and Michalski 2017). In order to capture these complexities,

recent research has considered the unequal-area FLP (UA-FLP), trying to locate departments of varying dimensions in constrained space such that nonoverlap and realistic placement are guaranteed (Drira et al. 2007).

Furthermore, the feasibility of the layout is also influenced by physical constraints, such as fixed infrastructure, safety clearances, and transportation pathways. Such factors are of prime importance in real-world environments and can play a major role in influencing material routing and department location. For example, obstacles in the form of walls or pillars can limit viable solutions and make internal logistics more complex, especially when material handling is vehicle-based using forklifts or automated guided vehicles (AGVs) (Besbes et al. 2020). Along with these complexities, real production environments experience high operational variability. Remanufacturing environments, for example, experience significant processing-time variability due to uncertainty in core quality and condition, as demonstrated by Kekre et al. (2003). Material flows among departments may vary according to the type of product, season, or availability of resources, whereas dynamic layout models attempt to modify and evolve over time. However, such reconfigurations are often expensive and disruptive, leading to production downtime (Anjos and Vieira 2017, Hosseini-Nasab et al. 2018, Kikolski et al. 2018). This challenge has been observed in studies of next-generation factory layouts. Benjaafar et al. (2002) argued that facilities need layouts designed for robustness when both production uncertainty and relayout costs are high, rather than layouts optimized for a single set of conditions. Instead of frequent time-based modifications, this research aims to create a layout that maintains consistent performance across a set of predefined disruption scenarios. The physical layout remains fixed, whereas flow routing and ramp selection adapt to each scenario. This scenario-based strategy enables long-term planning through the capture of essential operational variations without demanding repeated physical reconfigurations.

Uncertainty is a key issue in FLP modeling. Material flows in the real world vary based on seasonality, equipment breakdowns, scheduling changes, and supply unpredictability. Home appliance manufacturing industries that handle various product lines, process dependencies, and multimodal transport type are primarily subject to such uncertainty (Derakhshan Asl and Wong 2017). Disregarding these layout inconsistencies can lead to inefficient or unrealistic designs when applied to real-world environments. There have been some attempts to account for uncertainty through probabilistic, interval-based, or scenario-based approaches. For example, Peng et al. (2018) dealt with stochastic flows, and Yazdani et al. (2023) suggested robust optimization models based on uncertainty scenarios. Most of these models provide theoretical contributions but are not applied in practice. They tend to employ hypothetical scenarios and omit essential aspects or operational details. Moreover, few have used historical production data to estimate uncertainty or examine the robustness of the layout under real disruptions.

To fill these gaps, this article proposes a scenario-based robust optimization model that advances the standard UA-FLP by incorporating a simultaneous routing and layout logic. The model is a mixed-integer linear program (MILP) formulation that treats interblock connections not as infinite-capacity links, but as capacity-constrained bottlenecks (ramps) whose availability varies by scenario. By integrating soft constraints for capacity violations, the model prioritizes operational feasibility under stress conditions, a critical feature for industrial realism often missing in rigid academic formulations.

The proposed approach is validated through a case study of a leading European home appliance manufacturer. We focused on the washing machine plant, where departments such as plastic injection, metal forming, painting, and final assembly are physically distributed among blocks. Material flow between blocks occurs through designated ramps, subject to capacity and availability constraints.

The main contributions of this research are threefold.

- **Methodological Integration of Layout and Routing:** We present a formulation that solves the facility layout problem and the interblock network flow problem simultaneously. This allows the model to optimize department locations based on the availability and capacity of specific transport ramps across different scenarios, providing a higher degree of realism than distance-based metrics alone.
- **Feasibility-Robust Formulation:** Unlike traditional robust models that may become infeasible under severe disruptions, our model incorporates a penalty-based soft constraint mechanism. This allows the system to identify the "least-disruptive" feasible layout even when scenarios exceed nominal design capacities (e.g., during combined ramp failures).
- **Data-Driven Validation Framework:** We demonstrate a systematic framework for converting raw historical maintenance and production logs into discrete disruption scenarios. This bridges the gap between theoretical uncertainty modeling and practical industrial data availability, validated via simulation to quantify transport efficiency gains.

The remainder of this paper is structured as follows. The Literature Review section surveys the literature. The Model section presents the mathematical framework. The application section explains the case study and development of the scenario. The Results section reports the computational results and findings. Finally, the Conclusion section concludes the paper and suggests future research.

Related Literature

Facility layout design is a classic industrial engineering problem that deals with the arrangement of departments, machines, or workstations to enhance efficiency and minimize material handling costs. Classical models generally presume static layouts, deterministic data, and single-floor settings. The increased sophistication of contemporary manufacturing—marked by flexible systems, dynamic flows, and environmental concerns—has prompted researchers to examine more realistic problems with dynamic, uncertain, and multifloor settings. These have led to new modeling directions including dynamic facility layout problems, robust and stochastic formulations, and multiobjective models addressing cost, energy, and space utilization.

In the literature, several papers have reviewed developments on the facility layout problem. Drira et al. (2007) typified layout issues by space configurations, decision variables, and solution approaches. Ahmadi et al. (2017) highlighted modeling complications of multifloor settings, including elevator design and interfloor flows. Pérez-Gosende et al. (2021) surveyed the growing application of metaheuristics and hybrid optimization techniques in practical layouts. Building on these foundational reviews, this paper focuses on optimization-based, robust and dynamic, simulation-supported, and application-specific facility layout models. Special attention is paid to research addressing uncertainty, accommodating operational variability, and striving for practical applicability in complicated manufacturing systems.

Optimization-Based and Metaheuristic Methods

A significant portion of the FLP literature has concentrated on optimization techniques, both mathematical programming and metaheuristics. These techniques address increasingly difficult layout issues—like unequal-area departments, energy-efficient layouts, multifloor layouts, or dynamic redesigns—while enhancing efficiency, reducing costs, or optimizing space.

Numerous studies have applied mathematical modeling and hybrid heuristics to enhance layout decision-making. Dorrah and Marzouk (2021) combined agent-based modeling, queuing theory, and a multiobjective assignment model for administrative facilities. Zhang and Lin (2024) applied multiagent reinforcement learning with 3D A* search in connector assembly, whereas Subulan et al. (2023) proposed a heuristic decomposition model including machine capabilities for the UA-FLP. Other metaheuristics include learning-based simulated annealing (Lin et al. 2024), a modified Non-dominated Sorting Genetic Algorithm II for an air conditioner plant (Guo et al. 2023), and teaching-learning–based optimization for through-aisle layouts (Zhang et al. 2024).

Recent studies have incorporated sustainability and energy efficiency into facility layout modeling. Zhang et al. (2022) proposed an energy-conscious model using multiobjective particle swarm optimization, and Peron et al. (2020) developed dynamic layout planning via three-dimensional (3D) mapping and motion capture. Mayadunne et al. (2024) presented a two-stage MILP for warehouse storage efficiency and bin sizing. Other niche methods include Monarch butterfly optimization for AGV-based single-loop layouts (Kim and Chae 2019), discrete multiobjective particle swarm optimization in multiworkshop settings (Guan et al. 2019), and differential evolution in double-row layouts (Ji et al. 2023).

Reinforcement learning and hybrid metaheuristics have been considered for their flexibility under uncertainty. Kaven et al. (2024) utilized multiagent reinforcement learning with proximal policy optimization in flexible layouts, and Klar et al. (2022) demonstrated enhancements in large-scale layouts with double deep Q network. Configuration-space and slicing-based encodings were developed in Liu et al. (2020) and La Scalia et al. (2019). Fuzzy programming and robust models were presented in Kaveh et al. (2014) and Saeidi (2024), whereas simulation-supported optimization appeared in Emami and Nookabadi (2013) and Jolai et al. (2012). Fuzzy models combined with metaheuristics were used in Abedzadeh et al. (2013), Zhun et al. (2021), and Besbes et al. (2021). Layout models were also extended with domain-specific constraints and multiobjective performance, such as double-row MILPs with valid inequalities (Amaral 2024), I/O- constrained UA-FLPs (Muhayat and Utamima 2024), and mixed-integer nonlinear formulations (Seyedi et al. 2024). Multifloor or multiobjective layouts were studied using ant colony (Liu and Liu 2019) or permutation-based Genetic Algorithms (GAs) (Kalita and Datta 2024). Other contributions include robust layout enhancements by power values with normal search Abedzadeh et al. (2013), hybrid simulated annealing approaches (Matai 2015, Oktaviani and Utamima 2024), and simulation-integrated demand forecasting using long short-term memory (Shoushtari et al. 2024). Collectively, these works show that metaheuristic and mathematical models continue to advance toward more realistic layout problems. Nonetheless, they generally do not address scenario-based robustness, flow uncertainty, or spatial constraints like interblock routing and ramp limitations, which this research seeks to address.

Dynamic and Robust Layout Models

Robust manufacturing layouts are needed to remain effective under unforeseen conditions such as fluctuating demand, supply disruptions, or shifting product flows. Frequent reconfigurations are costly, and disruptive, which motivates the use of strong and flexible facility layout models that change internally and maintain performance across multiple scenarios. A classification framework for layout design based on production uncertainty and relayout cost was developed by Benjaafar

et al. (2002). For scenarios with high reconfiguration costs, they advocated for robust layouts capable of acceptable performance under each scenario rather than optimization for a single condition. This scenario-based robustness concept forms the foundation for modern approaches to layout design under uncertainty.

Several studies have implemented scenario-based approaches specifically for facility layout modeling. Krishnan et al. (2008) presented a genetic algorithm-based model under different production scenarios, offering flexibility and lower costs than static models. Scenario-based models for safety were proposed by Jung et al. (2010) and Díaz-Ovalle et al. (2010), optimizing facility location under toxic vapor leak conditions to reduce risks. To address unpredictable material movements, Şahinkoç and Bilge (2018) used a quadratic assignment problem–based model with genetic algorithms, whereas Reisinger et al. (2022) proposed a multiobjective evolutionary process for flexible facility scenarios. Shoushtari et al. (2024) combined machine learning–based demand forecasting with scenario-based genetic optimization to improve layout adaptability and sustainability.

Broader robust optimization methods were later suggested for adaptive uncertainty handling. Pourvaziri et al. (2021) proposed a robust MILP model with a hybrid genetic-Tabu search algorithm, ensuring internal routing adaptations for layout stability without repeated physical reconfigurations. Similarly, Cheng et al. (2021) used a two-stage robust model with column-and-constraint generation to determine facility allocations for disruption risk mitigation. Xiao et al. (2019) optimized unequal-area dynamic layouts using improved particle swarm optimization to reduce rearrangement costs. In the healthcare sector, Tongur et al. (2020) demonstrated robust optimization with metaheuristics for hospital layouts with variable patient and staff movements, achieving significant efficiency improvement.

Multilevel and spatial issues were considered by Izadinia et al. (2014), who applied the robust price and inventory model by Bertsimas and Sim (2004) to multifloor installations without frequent physical updates. Stochastic demand was handled by Moslemipour et al. (2017) using a quadratic assignment model with simulated annealing to reduce sensitivity to demand variations. Peng et al. (2018) applied robust inventory and pricing with Monte Carlo simulation and genetic algorithms, incorporating transportation planning with dynamic facility layout design to enhance dependability under variable demands.

Although these works extensively advance layout modeling, none combine scenario-based robustness with interblock routing and spatial limitations like ramp capacities—a gap this paper addresses. By creating a scenario-based robust MILP model with these realistic industrial operating constraints, this work closes the gap between theoretical research and practical industrial applications in layout design under uncertainty.

Simulation-Based Approaches

Simulation approaches have become increasingly valuable for facility layout design under uncertain and variable conditions. Combining simulation with optimization, machine learning, or reinforcement learning allows scholars to evaluate layout performance under varying conditions and enhance decision-making flexibility.

Recent studies have explored simulation-integrated layout design using advanced algorithms and hybrid frameworks. For instance, Bi et al. (2024) proposed an optimization approach for workshop layouts using the sparrow search algorithm (SSA), aiming to reduce transportation costs and improve nonlogistic relationships. Their results, verified via FlexSim, showed that SSA provided stable and optimal solutions. Likewise, Klar et al. (2024) developed a transferable framework for multiobjective layout planning using discrete event simulation and deep reinforcement learning, which proved efficient for uncertain planning with transfer learning. In another study, Zúñiga et al. (2020) introduced a five-phase simulation and optimization framework for redesigning facility layouts, leading to more usable and resilient arrangements in industrial cases. Choi and Kim (2024) proposed an intelligent arrangement system combining MILP, simulation-based optimization, and digital twin technologies to refine layouts and reduce time and effort for efficient design.

Modeling realistic spatial limitations, Besbes et al. (2020) integrated the A* pathfinding algorithm with simulation and a genetic algorithm for more real-world-like arrangements, introducing movement logic into layout decision making. In modular construction, Yang and Lu (2023) contrasted arrangement configurations using simulation-based optimization and revealed cost-productivity tradeoffs. Safety was considered by Vasudevan and Son (2011), who used a multiparadigm simulation framework to analyze productivity and emergency evacuation, demonstrating that some layouts could achieve both. Lee (2012) combined an MILP model with simulation logic and preprocessing to improve computational efficiency and solution quality. These studies demonstrate that simulation enables robust layout assessment under uncertainty, spatial constraints, and time-varying flow, and its integration with optimization and learning enhances flexibility for industrial applications. A summary of robust facility location papers is provided in Table 1.

Although metaheuristics and reinforcement learning have shown promise for facility layout optimization, this research employs MILP because it naturally accommodates the multiscenario structure of the problem. MILP allows for explicit modeling of a fixed layout

Table 1. Summary of Robust Facility Layout Literature

Study	Solution method	Uncertainty method
Benjaafar et al. (2002)	Conceptual	Scenario based
Krishnan et al. (2008)	Metaheuristic	Scenario based
Jung et al. (2010)	MILP	Scenario based
Díaz-Ovalle et al. (2010)	MILP	Scenario based
Şahinkoç and Bilge (2018)	Metaheuristic	Stochastic
Moslemipour et al. (2017)	Metaheuristic	Stochastic
Besbes et al. (2021)	Metaheuristic	Stochastic
Xiao et al. (2019)	Metaheuristic	Dynamic
Tongur et al. (2020)	Metaheuristic	Robust
Peng et al. (2018)	Simulation	Stochastic
Pourvaziri et al. (2021)	MILP + metaheuristic	Robust
Cheng et al. (2021)	MILP	Two-stage robust
Reisinger et al. (2022)	Metaheuristic	Scenario based
Shoushtari et al. (2024)	ML + metaheuristic	Scenario based

Note. MILP, mixed-integer linear programming; ML, machine learning.

that must perform well across multiple disruption scenarios simultaneously, with scenario-specific flow routing decisions optimized within a unified framework. This approach provides deterministic solutions with known optimality bounds and enables transparent tradeoff analysis between expected cost, layout stability, and constraint violations—critical requirements for long-term industrial facility planning.

Although the facility layout problem has been studied in detail, there are still important gaps. Most models focus on specific issues such as unequal-area departments, changing demand, or uncertainty, but few combine these in a clear and practical way. Very few consider more than one spatial or operational constraint, such as interblock flows or low-capacity ramps. Scenario-based models are becoming more common but are often limited to simplified cases and are rarely tested with real manufacturing data. Models that consider flow changes or space constraints often cannot scale to large systems or provide practical solutions.

To address these problems, we present a scenario-based facility layout model designed to reflect realistic operating conditions. Using real flow and layout data from a home appliance factory, the model is solved across multiple disruption scenarios based on historical records. Although scenario-based robust optimization for facility layout has gained attention in recent literature, systematic scenario generation from disaggregated historical production data remains underexplored. Our approach extracts disruption characteristics—including failure frequencies, demand variations, and capacity constraints—from three years of operational records combined with expert validation, grounding the optimization model in observed patterns rather than theoretical assumptions. It also includes spatial constraints like interblock ramp limits and additional cost effects. By solving one large MILP model for all scenarios at once, the approach provides layout solutions that work well in normal conditions and remain reliable during disruptions. This study offers a practical and robust tool to support layout planning in real manufacturing environments.

Mathematical Model

This section presents the single-period robust facility layout model to address uncertainty in flow volumes, transportation costs, and ramp capacities. The model makes two types of decisions: **strategic design decisions** (department locations and block assignments) that remain permanently fixed across all scenarios and **tactical operational decisions** (flow routing and ramp usage) that adapt internally to each scenario's conditions without requiring physical reconfiguration. This formulation is appropriate for greenfield facility design or major brownfield redesign projects where the spatial layout represents a long-term commitment, and operational flexibility is achieved through intelligent routing rather than costly physical rearrangements.

The objective function includes three components: (i) expected transportation cost across scenarios; (ii) deviation penalties to ensure layout stability; and (iii) soft penalties for violations such as ramp capacity overruns. Two parameters regulate the tradeoffs: the deviation penalty weight controls how much the model values cost stability versus expected cost, whereas the violation penalty weight represents the cost of capacity overruns (e.g., overtime, expedited handling, or temporary capacity expansion). This formulation ensures that the resulting layout is not only cost-effective under expected conditions but also remains robust and functional when disruptions occur, avoiding designs that perform well on average but fail catastrophically in certain scenarios.

Transportation costs include internal movement within blocks (based on Manhattan distances and flow volumes), external transfers between blocks via ramps (based on ramp-to-ramp distances and scenario-specific cost factors), and closeness penalties for separating departments that should remain adjacent due to workflow or operational requirements. The model enforces several operational constraints that vary by scenario. Material flows between departments in different blocks must be routed through designated ramp pairs, whereas flows within the same block incur only internal transport costs. Each ramp has a capacity limit that can be exceeded at a penalty cost, allowing the model to find feasible solutions even when all scenarios cannot be accommodated within strict capacity bounds. The model tracks which ramps are actually used for each flow path, ensuring that routing decisions reflect real material handling requirements rather than artificial shortcuts.

The model ensures geometric feasibility through several spatial constraints. Departments within the same block are positioned such that they do not overlap, with

their internal distances (calculated as Manhattan distances) used for cost calculations and closeness penalties. Each department is assigned to exactly one block, and departments must fit entirely within their assigned block boundaries based on their dimensions. To encourage compact layouts and facilitate solution interpretation, blocks are filled sequentially, meaning lower-numbered blocks must be occupied before higher-numbered blocks can be used. Finally, the model enforces logical consistency in department positioning through symmetry and transitivity constraints, ensuring that relative positions between departments are coherent and geometric conflicts are avoided.

The complete mathematical formulation is presented in Appendix A, including the objective function (Equation (A.1)), scenario-dependent flow and capacity constraints (Equations (A.9)–(A.14)), spatial layout constraints (Equations (A.15)–(A.22)), block assignment and sequencing logic (Equations (A.23)–(A.30)), and precedence consistency constraints (Equations (A.31)–(A.32)).

Application: Real-World Facility Layout at a Home Appliance Plant

The suggested scenario-based robust facility layout design model is implemented in production plant of one of the largest producers of home appliances in Europe. The factory manufactures various products such as refrigerators, dishwashers, washing machines, dryers, ovens, and cooling units. The plant is a high-volume facility with stringent efficiency demands and contains many departments performing metal forming, coating, subassembly, final assembly, and testing. These departments are spread over physically distinct layout blocks, and hence the design issue is especially difficult because of interblock material flows and space limitations.

A prominent feature of the facility is that it possesses nonuniform departmental area requirements and a dynamic production environment. Departments such as plastic injection and metal forming require large floor areas for heavy machinery, whereas final assembly and testing stations occupy smaller, more flexible spaces. Production scheduling must simultaneously accommodate multiple washing machine models with different processing sequences, creating competing demands for shared painting lines and material handling capacity. Not only are material flows between departments high density and high volume, but they are also sensitive to demand pattern changes, product mix variations, and sensitive to operational disruptions. The presence of multiple ramps connecting blocks further complicates the routing options under varying capacity and cost considerations. All these factors make the facility an excellent application case for the study of robust and flexible layout designs.

To represent realistic operating complexity, the layout model considers multiple future scenarios instead of one nominal or worst-case flow matrix. The scenarios represent possible variations in internal flows, external transport distances, and ramp availability. Each scenario is generated from historical production data and domain knowledge and is assigned a probability to represent its likelihood. The formulation enables the model to determine a layout that is feasible and cost-effective for a broad range of operating conditions, representing realistic tradeoffs between efficiency, stability, and robustness.

Scenario Definition and Disruption Modeling

In order to assess the proposed layout's performance in the face of uncertainty, 10 discrete scenarios were established, each of which corresponds to a conceivable future state of the production system. The scenarios were created to embody common disruptions witnessed in industrial environments, such as flow variations, ramp outages, shifts in demand, and higher transportation expenses. This structured framework allows the model to process a broad spectrum of operating conditions.

The scenarios considered in this study were defined after a detailed analysis of historical production data and consultation with production experts. The data are composed of disaggregated daily and shift-level information for material flows among departments, recording seasonal fluctuations, annual growth, planned shutdowns, and statutory holidays. We reviewed these data to identify normal patterns and rare events, such as sudden increases in demand, slowdowns, supply interruptions, and ramp failures. For events that were not frequent in the observed data, their frequency and impact were defined based on maintenance records and expert judgment. In this way, each scenario in the model has either direct empirical support from the operational data or the practical experience of specialists in the manufacturing environment. This ensures that the testing of layout performance is conducted with realistic scenarios and covers the major sources of uncertainty faced by decision makers. The definition of these scenarios involved an iterative process. Initial model runs based purely on theoretical failure rates were reviewed by plant managers; based on their feedback, the probability and impact of the Combined Stress scenario were refined to better align with the frequency of actual historical maintenance incidents, ensuring the final results represented actionable operational risks.

Ten disruption scenarios were formulated from expert opinion and historical data to capture a variety of real-world operating conditions. Scenario 1 is the base case with nominal flow volumes and all ramps available (25%). Scenario 2 assumes a moderate escalation of critical flows (*peak demand*, 15%), and Scenario 3 simulates a uniform decrease in total flow (*low demand*, 15%). Scenario 4 models the failure of one ramp (*ramp 2 failure*, 10%), and Scenario 5 assumes decreased output in selected key departments (*production disruption*, 8%).

Scenario 6 blends greater flow demand with a ramp failure (*combined stress*, 8%). Scenario 6 (Combined Stress) explicitly models co-occurrence of peak demand and ramp failure, representing the most operationally critical compound disruption based on expert assessment and historical incident analysis. Scenario 7 depicts a localized shortage of incoming materials (*supply shortage*, 6%). Scenario 8 postulates a greater system throughput requirement (*future expansion*, 8%), Scenario 9 assumes a large increase in interblock transportation costs (*energy crisis*, 3%), and Scenario 10 considers additional flow resulting from rework across all units (*quality problems*, 2%). A probability was associated with each scenario to reflect its practical likelihood and guide the scenario-based optimization process. These scenarios reflect operational variability patterns identified through analysis of historical production data and consultation with plant managers, capturing the range of conditions the facility experiences during normal operations and disruption events.

Each scenario is defined by a particular flow matrix p_{ij}^{ω}, ramp capacity vector R_r^{ω}, and, where applicable, external transport cost coefficient c_{ω}^{ext}. Flow data for Scenario 1 are extracted directly from actual plant records, whereas the remaining scenarios were constructed by applying predefined modification rules to the base flow matrix. Scenario 2 models a surge in production demand for specific operations, Scenario 3 imposes an across-the-board 30% reduction to reflect a planned slow-down, and Scenario 9 alters the cost parameters on interblock transportation by tripling c_{ω}^{ext}, discouraging transfers when energy prices rise or carbon constraints tighten. In disruption cases such as equipment failure, ramp capacities were set to zero. The capacity of ramps between blocks varied by scenario. In Scenarios 4 and 6, Ramp 2 is completely closed, and the model must reroute flows through the remaining ramps. Two optimization parameters were robustly set.

Computational Results

Within this section, we present the computational results of the solution of the proposed scenario-based robust facility layout model. We focus on the performance of the layout, cost structure, and robustness in the defined scenarios. All experiments were conducted using actual data of the above-presented case study. Table 2 summarizes scenario parameters and characteristics.

The scenario-based robust facility layout problem was solved with Gurobi 12.0.1 on a 12-core processor with multithreading. The optimization took 1,650.34 seconds, and the final optimality gap was less than 1%.

The total objective value was 831,553.2 units and had three principal components: expected scenario cost, deviation penalty, and capacity violation penalty. Table 3 gives the numerical values and contributions of each term. The expected cost term ($\sum_s \pi_s z_s$), with 726,496.4 units (87.4%), is the weighted average operational cost across all scenarios by its probability. This term represents the overall performance of the fixed layout design under variable flow and ramping conditions.

The deviation penalty adds 77,086.7 units (9.3%). It penalizes uncertainty in scenario costs and encourages stable layout performance under uncertainty. This deviation penalty ensures consistent layout performance across scenarios rather than excelling in some while failing in others. This stability is critical for manufacturing managers, as layouts optimized only for average conditions might become severely inefficient during peak production or equipment failures, forcing costly adjustments like overtime material handling or emergency rerouting through congested pathways. The last element is the violation penalty ($w \cdot \sum v$), which is worth 27,970.2 units (3.4%), and it represents total ramp capacity exceedance over scenarios, based on the penalty multiplier. This term encourages the model to avoid infeasibility but permit soft constraint handling when absolutely required.

This decomposition emphasizes the tradeoff among cost efficiency, operational stability, and robustness in layout design. Expected cost is prioritized as much as possible, whereas deviation and violation penalties keep the layout resilient and realistically implementable over a broad spectrum of disruptions.

Figure 1 and Table 4 show the overall cost for each of the ten predefined disruption scenarios. The costs

Table 2. Scenario Parameters and Characteristics

Scenario	Probability (%)	Ramp status	Description
Base case	25	All active	Nominal flow volumes
Peak demand	15	All active	40% increase in critical flows
Low demand	15	All active	Uniform 30% flow reduction
Ramp 2 failure	10	R2 = 0	Infrastructure failure
Production disruption	8	All active	30% capacity in key production areas
Combined stress	8	R2 = 0	Peak demand + ramp failure
Supply shortage	6	All active	50% drop in input flows
Future expansion	8	All active	30% throughput expansion
Energy crisis	3	All active	Tripled external transport cost
Increased rework	2	All active	20% rework-induced inefficiency

Notes. Probabilities reflect scenario likelihoods based on historical data and expert judgment. All parameters sum to 100%.

Table 3. Objective Function Components

Component	Value	Share (%)
Expected cost ($\sum_\omega \pi^\omega C^\omega$)	726,496.4	87.4%
Deviation penalty ($\lambda \cdot \sigma$)	77,086.7	9.3%
Violation penalty ($w \cdot \sum v$)	27,970.2	3.4%
Total objective	831,553.2	100%

Notes. Values are in cost units. Shares indicate the percentage contribution to the total objective. $\sigma = \sum_{\omega \in \Omega} \pi^\omega (\delta^{+,\omega} + \delta^{-,\omega})$.

indicate how the final robust layout operates under varying flow configurations, ramp capacities, and external cost levels. The layout is kept constant for all the scenarios, and only flow routing and ramp utilization change according to scenario-dependent parameters. The list of departments is listed in Table 5.

The minimum cost occurs in the *Production Disruption* case (286,184.2 units), followed by *Low Demand* (497,033.0 units), both of which are decreased levels of operation. The counterintuitively low cost in the *Production Disruption* scenario arises because this scenario models a partial production shutdown with only 30% capacity in key departments, resulting in substantially reduced material flow volumes (1,890 units total interblock flow, as shown later in Table 8). Although operationally undesirable due to lost production, the reduced material movement translates directly to lower transportation costs within the layout optimization model. This highlights an important distinction: the model minimizes material handling costs for a given flow pattern, not production output or revenue. In practice, production managers would seek to avoid such disruptions

Table 4. Scenario-Wise Total Costs

Scenario	Total cost
Production disruption	286,184.2
Low demand	497,033.0
Base case	710,027.0
Ramp 2 failure	710,027.0
Supply shortage	698,538.0
Quality problems	852,023.0
Future expansion	923,021.0
Peak demand	951,764.0
Combined stress	951,764.0
Energy crisis	961,251.0

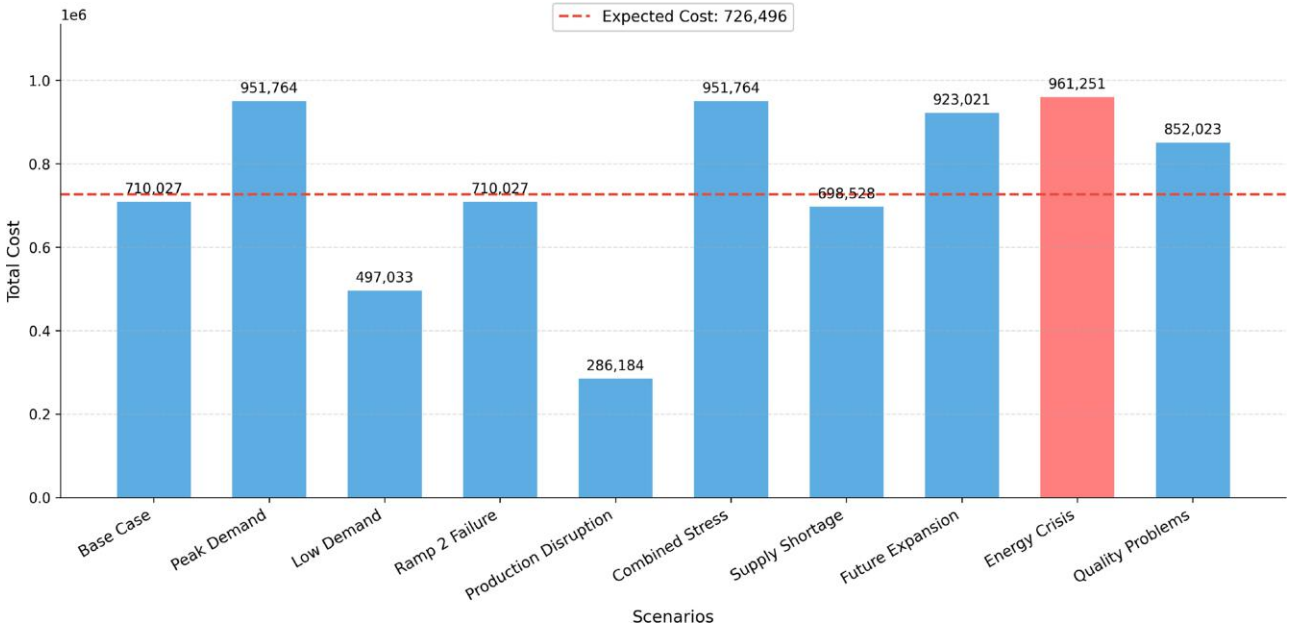
Note. The layout performs cheapest under *Production Disruption* and *Low Demand*, remains stable in *Ramp 2 Failure*, and reaches its highest costs in *Combined Stress* and *Energy Crisis*.

despite their lower handling costs, as the revenue losses far exceed any transportation savings. The *Base Case* and *Ramp 2 Failure* cases both have the same cost of 710,027.0 units, meaning that the layout has the capacity to absorb a ramp failure at no extra cost.

Medium-range costs are found in *Supply Shortage* (698,538.0) and *Quality Problems* (852,023.0), where flow disruptions are isolated. Higher costs are incurred in *Future Expansion* (923,021.0) and *Peak Demand* (951,764.0) because of high flow volumes and near-capacity ramp utilization. The most costly scenarios are *Combined Stress* (951,764.0) and *Energy Crisis* (961,251.0), involving several disruptions such as ramp failures and high external costs.

Overall, the scenario spread showcases the model's capacity to accommodate a broad set of disruptions. The

Figure 1. (Color online) Scenario-Wise Total Cost Comparison



Note. The figure illustrates how total costs vary across disruption scenarios while the layout remains fixed.

Table 5. Department Reference: ID to Area Description

ID	Area description
1	Mainplant Semi Product Whse.
2	Plastic Warehouse
3	Plastic Raw Material Whse.
4	Mechanic Warehouse
5	Export Goods Warehouse
6	Packaging Warehouse
7	Pre Assembly Line
8	Assembly Line
9	Final Assembly Line
10	Foamed Door Line
11	Packaging Line
12	Compressor Warehouse
13	Foamed Body Line
14	Door Glass Warehouse
15	Gasket Raw Material Whse.
16	Pipe Roll Warehouse
17	Plastic Sheet Whse./Extruder
18	CTF
19	DTF
20	Fin Warehouse
21	Other Production

Notes. Department identifiers used in Figures 5 and 6. Critical (high-volume) departments are highlighted in the layout figures.

cost spread between the worst and best scenarios amounts to almost 70%, illustrating the benefit of robust planning.

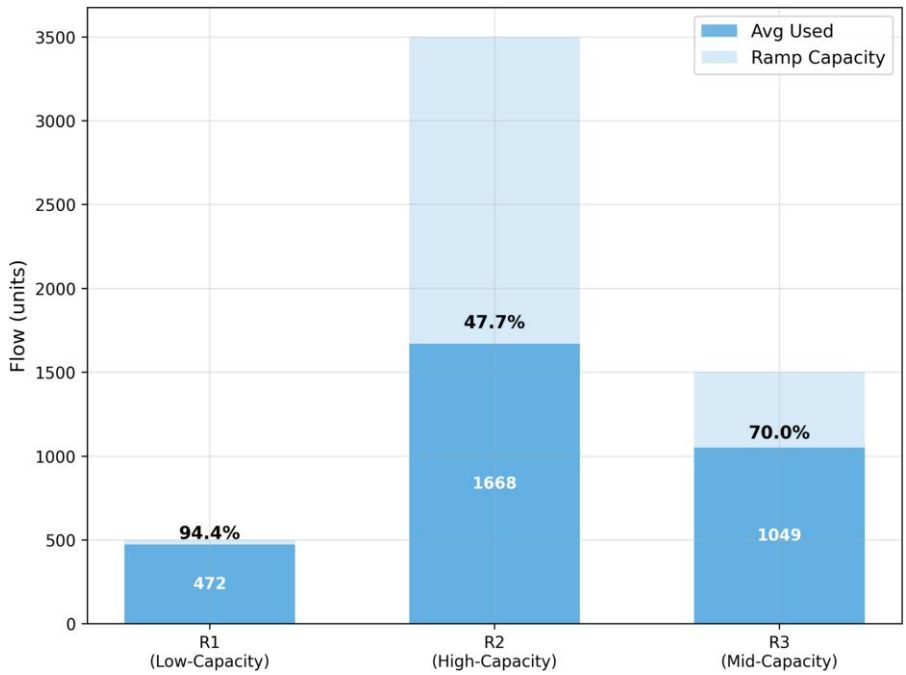
Figure 2 gives the average utilization of the ramps over all the scenarios, weighted by scenario probabilities. Ramp R1 has the highest relative utilization of 94.4%, which corresponds to an average of 236.1 units of flow. Although it has lower capacity, R1 is busy in the majority of the scenarios as it is involved in short inter-block transfers. Ramp R2 has the highest volume (1,669 units in total) but has only 47.7% utilization because of its high capacity. It is the default route for large flows but has sufficient buffer to accommodate peak or stress loads. Ramp R3 has 70.0% utilization, frequently being used as a flexible backup when R1 or R2 are congested.

Figure 3 illustrates ramp usage by scenario. Ramp R2 dominates high-flow scenarios such as *Peak Demand* and *Future Expansion*, but stays under capacity. Ramp R3 experience peaks in *Combined Stress* and *Production Disruption*, reflecting its use as overflow routes. Ramp R1 is used continuously, with a peak in usage for exceptional circumstances such as *Combined Stress*. In the *Ramp 2 Failure* scenario, R2 usage falls, whereas R1 and R3 compensate effectively, validating layout resilience.

Figure 4 provide a comprehensive overview of ramp capacity violations for all specified scenarios. Figure 4(a) illustrates the number of scenarios for which each ramp has a capacity violation. Interestingly, all three ramps (R1, R2, and R3) have violations in just 2 of the 10 scenarios—namely, the *Combined Stress* and *Ramp 2 Failure* instances.

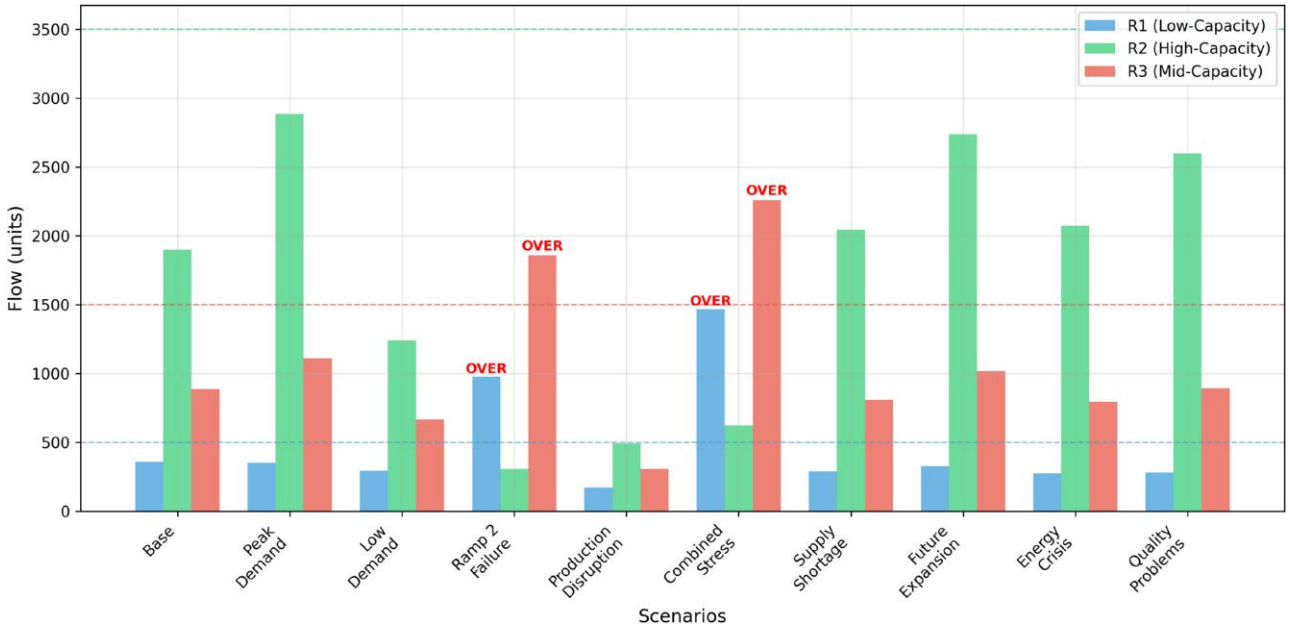
Figure 4(b) further disaggregates the total violation amount by scenario. Violations are mostly localized in the most extreme disruption scenarios: the *Combined Stress* scenario incurs the highest cumulative violation (2,224 units), followed by *Ramp 2 Failure* (1,017 units). All

Figure 2. (Color online) Average Ramp Utilization (Weighted Across Scenarios)



Note. Ramp R1 is the busiest due to frequent short transfers, R2 carries the largest volume but with spare capacity, and R3 serves as a flexible backup.

Figure 3. (Color online) Ramp Utilization by Scenario

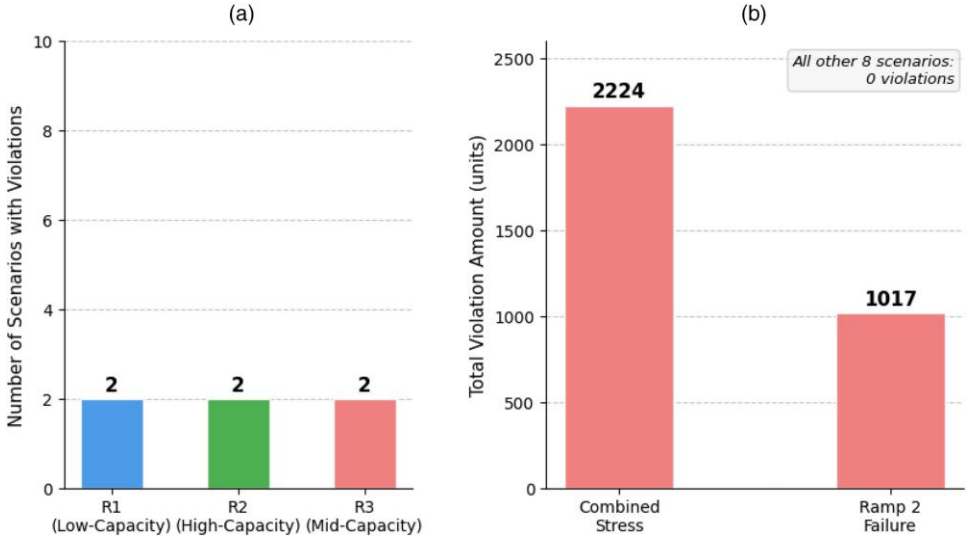


Note. R2 dominates in high-flow cases, R3 absorbs stress peaks, and R1 remains continuously active, with balanced compensation during Ramp 2 failure.

remaining scenarios, including those pertaining to demand fluctuations, quality issues, and supply disruptions, result in either zero or negligible violations. This trend indicates that the model is most sensitive to compound or ramp-specific disruptions, which can drive one or more ramps outside of their design specifications. By contrast, normal fluctuations are contained within available capacities.

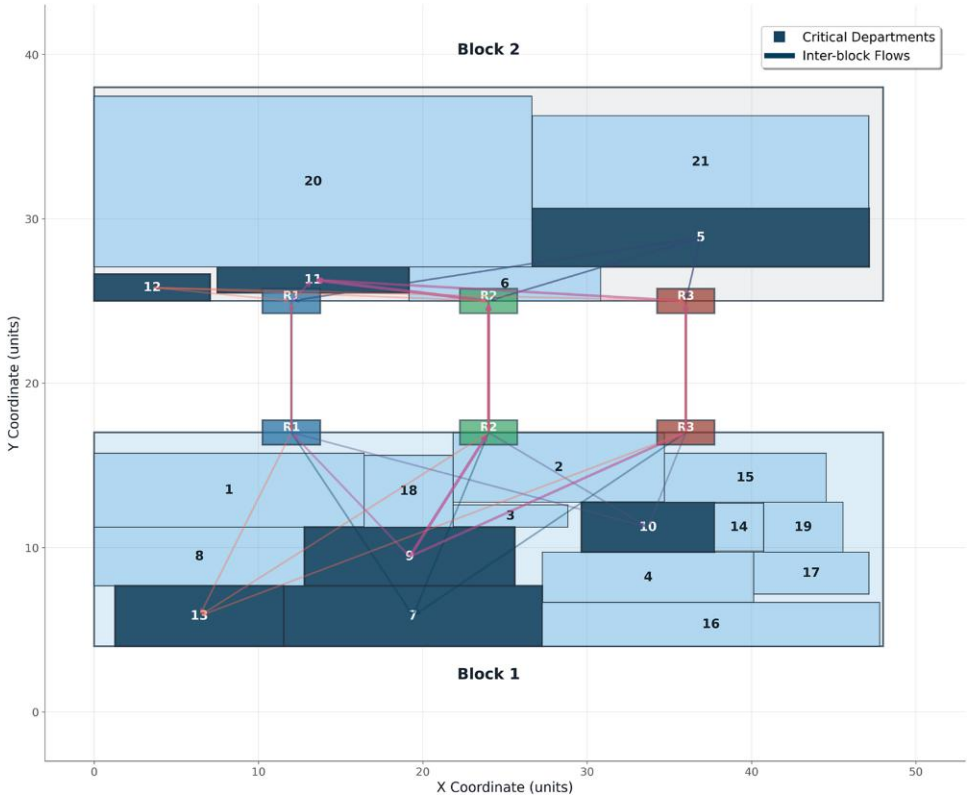
The violation statistics for each ramp reveal that although all ramps are violated in the same number of scenarios (two each), they differ in the extent of violations: R1 is the most violated ramp with a total violation of 1,850 units, followed by R3 (1,354 units) and R2 (1,017 units). These variations capture both the starting ramp capacity values and the underlying flow distribution in the disrupted scenarios. The extensive violations for R1,

Figure 4. (Color online) Scenario-Based Violation Analysis: (a) Number of Scenarios with Ramp Capacity Violations and (b) Total Violation Amounts



Note. Violations occur only in the most severe cases, with peaks in *Combined Stress* and *Ramp 2 Failure*, whereas all other scenarios remain within capacity.

Figure 5. (Color online) Final Robust Facility Layout



Note. The layout clusters high-volume and high-risk departments near ramps, allowing flows to be flexibly split and rerouted under disruptions, achieving both efficiency and resilience.

specifically, underscore its sensitivity under compound stress conditions and indicate a likely bottleneck in the design. Operational managers validated this finding, noting that the Combined Stress scenario currently necessitates overtime labor in the actual facility, mirroring the model's reported violations. Consequently, rather than constraining the model to force unrealistic zero-violation solutions, the calculated violation magnitudes are used to dimension contingency budgets for these high-risk periods. In total, these findings validate the application of scenario-based robustness to facility layout design. Although the model succeeds in avoiding violations in most instances, selective refinement of ramp capacity or flow assignment for certain high-risk scenarios (like combined stress or ramp failure) would further increase the system's resilience. This study highlights the value of considering not just the frequency but also the severity of capacity violations in assessing layout robustness.

Optimal Layout and Flow Behavior

Figure 5 shows the final robust facility arrangement for the scenario-based MILP model presented in this study, with department identifiers listed in Table 5. It integrates disruptions from all scenarios to achieve responsiveness and stability. The arrangement positions 21 departments in two blocks with only four active interblock flows after optimization, achieving low external transport cost. The block assignments for the departments are shown in Table 6.

The model determines ramp utilization patterns for each scenario during optimization, identifying a layout that accommodates varying flow demands and ramp capacities across different operational conditions. Rather than strict balance, the model enables some ramps—like Ramp 2—to carry a greater proportion of high-volume flows as required. Seven high-volume or high-risk departments have been strategically located in the two-block configuration to minimize disruption and enable efficient movement between blocks. Clustering high-interaction departments close to the ramps also enables the system to easily redirect flows in response to changing conditions.

In Block 2, Departments 11 and 6 are located adjacent to all three ramps and are thus important points for the transfer of material between blocks. This arrangement accommodates significant outbound flows to Block 1, particularly to Department 9, which is situated near its own ramps. Other major flows, such as those involving Departments 5 and 12, also move through these gateways as circumstances dictate. Large flows are split over several ramps, which helps ensure both efficiency and redundancy. Under normal circumstances, flows such

Table 6. Department-to-Block Assignments

Block	Departments assigned
Block 1	1, 2, 3, 4, 5, 6, 8, 10, 13, 14, 15, 16, 17, 18, 19
Block 2	7, 9, 11, 12, 20, 21

Note. The assignments show that Block 1 accommodates most departments, whereas Block 2 contains a smaller subset, reflecting the final distribution across blocks.

as from Departments 9 to 11 or from 5 to 7, are spread over a number of ramps to prevent congestion. When disruptions happen—such as a ramp failure or an unexpected shift in demand—the system responds by rerouting flows as necessary.

The general arrangement distinguishes clearly between stable and variable processes, locating stable departments in less busy areas and retaining more variable or higher-risk ones in closer proximity to principal access points. This modular approach enables the facility to respond well to various operational challenges. By strategically locating departments and utilizing ramps with flexibility, the arrangement realizes a pragmatic balance between efficiency, resilience, and flexibility.

The model accommodates flexible routing through the possibility of splitting flows across ramp pairs. Table 7 highlights the main multiramp flows. For example, one of the high-volume flows (5→7) utilizes all same-type ramp pairs (R1-R1, R2-R2, R3-R3) and switches to R1 and R3 under Ramp 2 failure. Flow 9→11 utilizes R1-R1 and R2-R2 and adapts under combined stress. Some flows such as 8→12 are invariant under all scenarios, whereas others experience redistributions under stress or peak demand, reflecting the model's flexibility.

Table 8 shows total interblock flow volumes per scenario. *Production Disruption* experiences the lowest flow (1,890 units), caused by reduced demand. *Low Demand* also yields a relatively small flow of 2,783 units. In contrast, *Energy Crisis* and *Combined Stress* scenarios generate the highest volumes (4,198 and 4,122 units), which are in line with capacity violations discussed earlier.

Base Case and *Ramp 2 Failure* scenarios provide the same flow (3,220 units), which means that the model compensates for Ramp 2 loss by redistribution without reducing overall transport amount. The variation in flow, from 1,890 to 4,198 units, demonstrates the robustness of the layout under different production and ramp conditions.

Table 7. Main Interblock Flows Using Multiple Ramps Across Scenarios

Flow ($i \rightarrow j$)	Ramps used	Consistency across scenarios
5 → 7	R1, R2, R3	Always multiramp
5 → 10	R1, R2, R3	Always multiramp
9 → 11	R1, R2, R3	R3 not used in some low flows
12 → 13	R1, R2, R3	Always multiramp

Note. The table highlights major interblock flows that consistently distribute across multiple ramps, indicating robust routing flexibility even under varying scenario conditions.

Table 8. Total Interblock Flow per Scenario

Scenario	Total interblock flow (units)
Production disruption	1,890
Low demand	2,783
Base case	3,220
Ramp 2 failure	3,220
Supply shortage	3,408
Quality problems	3,601
Future expansion	3,880
Peak demand	4,025
Combined stress	4,122
Energy crisis	4,198

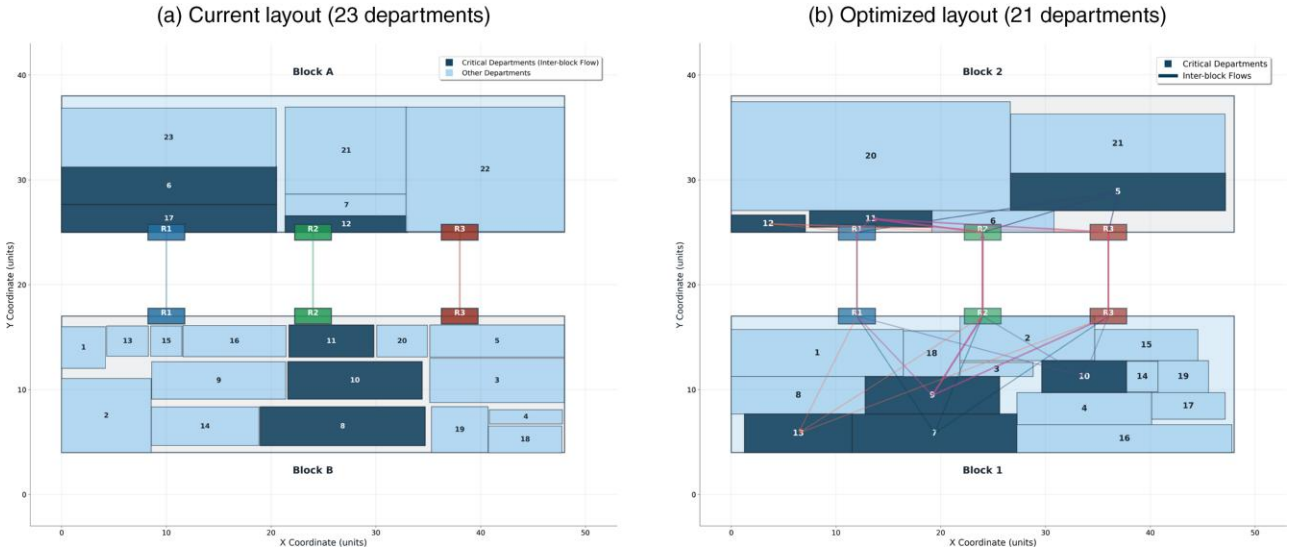
Note. Interblock flows increase steadily from reduced-operation cases to stress scenarios, with the lowest flow in *Production Disruption* and the highest in *Energy Crisis*.

Simulation Analysis

To validate the operational improvement achieved by the optimized facility layout, a simulation-based comparison was carried out using visTABLE 3.0.322, which is factory layout and material flow simulation software. Both the original layout—representing the plant's layout prior to optimization—and the optimized layout—achieved through the proposed scenario-based robust MILP model—were simulated within the same spatial boundaries. Logistics performance was quantified in terms of total transportation effort and space utilization. Prior to optimization, several managerial consolidation decisions were made to better reflect functional groupings. Specifically, the two Mainplant Semi Product Warehouses were merged into one, and the two plastic and two final warehouses were each combined, reducing the total from 23 to 21 departments in the optimized layout. Figure 6 presents a comparison of the current and optimized layouts.

For transportation efforts, the simulation quantified the total traveled distance for all intralogistics activities, disaggregated by mode of transport: conveyors, forklifts/AGVs, and manual handling. As can be seen in Table 9, the optimized layout reduced the overall transport distance from 233.02 km to 188.17 km within the operational period, an overall decrease of 19.3%. Significant improvements were recorded in all transport modes: Manual handling was reduced by 64.7%, forklift/AGV travel by 23.9%, and conveyor usage by 16.8%. These improvements are attributable to the improved spatial proximity of interdependent departments and more effective ramp assignments identified by optimization. Although labor costs were not explicitly modeled, the reduction in manual transport distance serves as a proxy for workforce impact. This decrease

Figure 6. (Color online) Comparison of Current and Optimized Facility Layouts



Note. The optimized layout consolidates departments and repositions high-flow units closer to ramps, reducing interblock travel distances and improving spatial efficiency.

directly correlates with reduced operator fatigue and labor hours, demonstrating quantifiable gains in human productivity. Qualitatively, the simulation shows a clear structural change. In the original layout, busy departments were scattered near the building's edges, which forced materials to cross the center repeatedly. The optimized layout groups these high-volume departments right next to the ramps. This change replaces long, crossing paths with short, direct routes, greatly simplifying the flow of materials.

In addition to flow efficiency, space utilization was also evaluated. Both layouts were designed within the same total floor area of 1,333 m^2, but their internal allocation of production and open space differed significantly. As shown in Table 10 and Figure 7, the optimized layout increased the production area from 748.7 m^2 to 980.7 m^2 and reduced open space from 584.0 m^2 to 352.1 m^2. This represents a 31% increase in space dedicated to production activities, boosting the production area share from 56.2% to 73.6%.

Overall, these simulation results reinforce the practical value of the proposed layout model. By reducing material handling distances and enhancing space utilization, the optimized layout supports leaner internal logistics and more efficient use of production floor space. From a managerial perspective, such improvements contribute directly to lower operational costs, reduced congestion, and smoother production flow. More importantly, the scenario-based design approach offers decision-makers a resilient and forward-looking layout that remains effective under changing production conditions, enabling better long-term planning and reduced need for disruptive reconfigurations.

Table 9. Comparison of Logistics Performance Between Original and Optimized Layouts

Metric	Original layout	Optimized layout	Improvement
Total distance (km)	233.02	188.17	−19.3%
Conveyor distance (km)	177.78	147.98	−16.8%
Forklift/AGV distance (km)	50.82	38.63	−23.9%
Manual distance (km)	4.42	1.56	−64.7%

Note. The optimized layout reduces travel distances across all transport modes, with the largest relative improvement in manual handling and notable gains in forklift and conveyor movements.

Conclusions

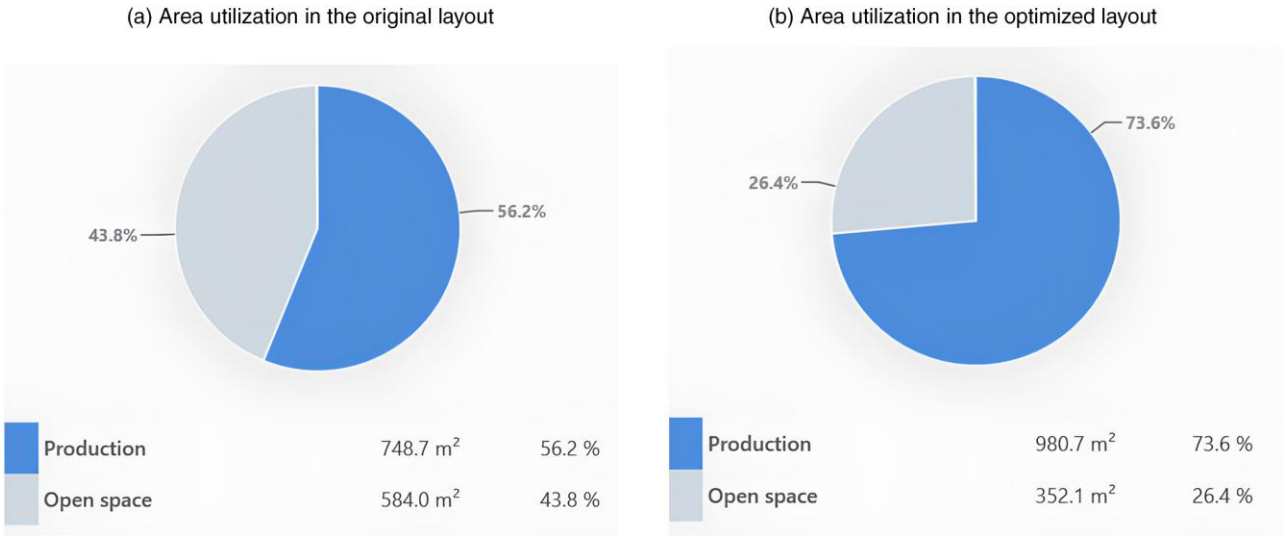
This research presented a scenario-based robust facility layout model to tackle the uncertainty and operational variability that are commonly found in real-world manufacturing practice. In contrast to deterministic models that optimize the layout with respect to a single nominal assumption, the proposed model integrated 10 disruption scenarios, each reflecting a realistic operational condition. Scenario probabilities were introduced to reflect the likelihood of every

Table 10. Comparison of Space Utilization Between Original and Optimized Layouts

Metric	Original layout	Optimized layout
Production area (m^2)	748.7	980.7
Open space (m^2)	584.0	352.1
Production share (%)	56.2%	73.6%

Note. The optimized layout increases production area and share while reducing unused open space, indicating more efficient utilization of available floor capacity.

Figure 7. (Color online) Comparison of Production and Open Space Share in Original and Optimized Layouts



Note. The optimized layout dedicates a greater share of the fixed floor area to production, reducing unused space and improving overall utilization.

disruption, enabling the model to provide a reliable layout in a broad spectrum of future states. Joint optimization of department placement, flow routing, and ramp utilization made the layout cost-effective even in the event of equipment breakdown, high demand, or limited capacity.

The results confirmed that the proposed layout effectively balanced spatial efficiency, transport cost, and robustness. Departments with high flow intensity were placed in such a manner as to restrict interblock traffic, and routing flexibility across a range of ramp types created the capacity to respond to stress without appreciable loss of performance. Although the average utilization of the ramps was within capacity, the scenario-based analysis revealed how stress conditions would create overloads on specific ramps, particularly when the key infrastructure was out of service. The model handled such occurrences by intelligent redistribution of flows, achieving operational feasibility for most scenarios and minimizing violations when unavoidable.

Significantly, the model was not premised on fixed routing or inflexible assignments. Rather, the model evaluates how flow paths would be routed under different disruption circumstances during the design phase, ensuring the layout remains feasible and efficient across operational scenarios. Scenario-by-scenario flow volume profiles also illustrated the model's capacity to accommodate different demand levels and supply bottlenecks without sacrificing transport efficiency.

Cumulatively, this research offers a robust and resilient solution to facility layout design under uncertainty. The model constitutes a rigorous decision support instrument for industrial planners that can anticipate and mitigate the impact of real-world disruptions. Through the combination of probabilistic modeling, ramp-level routing, and spatial constraints in a unified optimization framework, the study advances the state of layout planning in complex production systems. The major obstacles were in developing realistic disruption scenarios and managing computational intensity in solving a large-scale MILP. These were addressed through data preprocessing and careful scenario design so that the final model was still manageable but representative of industrial scenarios.

Future work can address integration into simulation environments, digital twins, or real-time layout adjustment based on live production data. The results of this research form a basis for such extensions and demonstrate the practical applicability of robust layout optimization in today's dynamic manufacturing environment.

From a managerial perspective, the model provides a systematic yet adaptive decision support instrument that allows layout planners to actively assess the impact of various disruption scenarios—be it equipment breakdowns, capacity changes, or demand spikes—on spatial assignments, interblock routing, and ramp usage. Rather than responding to disruptions after they have occurred, managers are able to utilize the model to create facility layouts that are resilient by design, with a guarantee of operational continuity and cost-effectiveness even in stressed situations.

Acknowledgments

The authors thank Professor Metin Türkay from Koç University for guidance and support throughout this research. The authors also gratefully acknowledge the contributions of Arçelik A.Ş.–Beko for collaboration.

Appendix A. Mathematical Model

Table A.1. Sets

Notation	Description
$i, j \in N$	Departments
$k, k_1, k_2, t \in \mathcal{M}$	Blocks
$r, r_1, r_2 \in R$	Ramps
$\omega \in \Omega$	Scenarios

Table A.2. Parameters

Notation	Description
l_i, w_i	Length/width of department i
$\bar{L}$, $\bar{W}$	Length/width of each block
p_{ij}^{ω}	Flow from department i to j in scenario ω
c_{ij}^{int}	Internal transportation cost from i to j
$c^{\text{ext},\omega}$	External transport cost in scenario ω
R_r^{ω}	Capacity of ramp r in scenario ω
$D_{r_1 r_2}$	Distance between ramps r_1 and r_2
κ_{ij}	Closeness rating between departments i and j
π^{ω}	Probability of scenario ω
λ	Weight of cost-deviation penalty
W	Weight of violation penalty

Note. $\bar{L}$ and $\bar{W}$ are fixed scalar parameters representing the identical dimensions for all blocks $k \in \mathcal{M}$.

Table A.3. Scenario-Independent (Design Variables) Decision Variables

Variable	Description
Y_{ik}	1 if department i is assigned to block k; 0 otherwise
T_k	1 if block k is used; 0 otherwise
x_{ik}, y_{ik}	Coordinates of department i in block k
u_{ijk}	1 if departments i and j are both in block k; 0 otherwise
$d_{ij}^{\text{int},x}$, $d_{ij}^{\text{int},y}$	Internal distance components between departments i and j
α_{ijk}, β_{ijk}	Nonoverlap direction indicators for departments i and j in block k (binary): (1,1): i precedes j horizontally; (0,0): j precedes i horizontally; (1,0): i precedes j vertically; (0,1): j precedes i vertically

Table A.4. Scenario-Dependent (Control Variables) Decision Variables

Variable	Description
$F_{ijk_1k_2r_1r_2}^{\omega}$	Flow from department i in block k_1 to department j in block k_2 via ramps r_1 and r_2 under scenario ω
$Z_{ijk_1k_2r_1r_2}^{\omega}$	1 if route from i in block k_1 to j in block k_2 uses ramps r_1 and r_2 in scenario ω; 0 otherwise
$d_{ij}^{\text{ext},\omega}$	External distance between departments i and j in scenario ω
v_{kr}^{ω}	Capacity violation at ramp r in block k under scenario ω
C^{ω}	Total transportation cost in scenario ω
$\delta^{+,\omega}$, $\delta^{-,\omega}$	Positive/negative deviation from expected cost in scenario ω

Table A.5. Auxiliary Variables

Variable	Description
$\bar{C}$	Expected total cost across all scenarios
V	Total weighted penalty for capacity violations

A.1. Objective Function

$$\min \bar{C} + \lambda \sum_{\omega \in \Omega} \pi^{\omega} (\delta^{+,\omega} + \delta^{-,\omega}) + w \cdot V, \tag{A.1}$$

where

$$\bar{C} = \sum_{\omega \in \Omega} \pi^{\omega} C^{\omega}, \tag{A.2}$$

$$C^{\omega} - \bar{C} = \delta^{+,\omega} - \delta^{-,\omega} \qquad \forall \omega \in \Omega, \tag{A.3}$$

$$V = \sum_{\omega \in \Omega} \pi^{\omega} \sum_{k \in \mathcal{M}} \sum_{r \in R} v_{kr}^{\omega}, \tag{A.4}$$

$$C^{\omega} = C_{\text{int}}^{\omega} + C_{\text{ext}}^{\omega} + P \qquad \forall \omega \in \Omega, \tag{A.5}$$

$$C_{\text{int}}^{\omega} = \sum_{i \in N} \sum_{\substack{j \in N \\ j \neq i}} p_{ij}^{\omega} c_{ij}^{\text{int}} (d_{ij}^{\text{int},x} + d_{ij}^{\text{int},y}), \tag{A.6}$$

$$C_{\text{ext}}^{\omega} = c^{\text{ext},\omega} \sum_{i \in N} \sum_{\substack{j \in N \\ j \neq i}} \sum_{\substack{k_1, k_2 \in \mathcal{M} \\ k_1 \neq k_2}} \sum_{r_1, r_2 \in R} F_{ijk_1k_2r_1r_2}^{\omega} D_{r_1 r_2}, \tag{A.7}$$

$$P = \sum_{i \in N} \sum_{\substack{j \in N \\ j \neq i}} \kappa_{ij} \left(1 - \sum_{k \in \mathcal{M}} u_{ijk}\right). \tag{A.8}$$

Subject to:

$$\sum_{\substack{k_1, k_2 \in \mathcal{M} \\ k_1 \neq k_2}} \sum_{r_1, r_2 \in R} F_{ijk_1k_2r_1r_2}^{\omega} = p_{ij}^{\omega} \left(1 - \sum_{k \in \mathcal{M}} u_{ijk}\right) \qquad \forall i, j \in N,\ i \neq j,\ \forall \omega \in \Omega, \tag{A.9}$$

$$F_{ijk_1k_2r_1r_2}^{\omega} \leq p_{ij}^{\omega} Y_{ik_1} Y_{jk_2} \qquad \forall i, j \in N,\ i \neq j,\ \forall \omega \in \Omega \qquad \forall k_1, k_2 \in \mathcal{M},\ k_1 \neq k_2,\ \forall r_1, r_2 \in R, \tag{A.10}$$

$$\sum_{i \in N} \sum_{\substack{j \in N \\ j \neq i}} \sum_{\substack{k' \in \mathcal{M} \\ k' \neq k}} \sum_{r' \in R} F_{ijkk'rr'}^{\omega} \leq R_r^{\omega} + v_{kr}^{\omega} \qquad \forall \omega \in \Omega,\ \forall k \in \mathcal{M},\ \forall r \in R, \tag{A.11}$$

$$F_{ijk_1k_2r_1r_2}^{\omega} \leq M Z_{ijk_1k_2r_1r_2}^{\omega} \qquad \forall i, j \in N,\ i \neq j,\ \forall \omega \in \Omega \qquad \forall k_1, k_2 \in \mathcal{M},\ k_1 \neq k_2,\ \forall r_1, r_2 \in R, \tag{A.12}$$

$$d_{ij}^{\text{ext},\omega} \leq M \left(1 - \sum_{k \in \mathcal{M}} u_{ijk}\right) \qquad \forall i, j \in N, i \neq j, \omega \in \Omega, \tag{A.13}$$

$$d_{ij}^{\text{ext},\omega} p_{ij}^{\omega} \left(1 - \sum_{k \in \mathcal{M}} u_{ijk}\right) \geq \sum_{\substack{k_1, k_2 \in \mathcal{M} \\ k_1 \neq k_2}} \sum_{r_1, r_2 \in R} F_{ijk_1k_2r_1r_2}^{\omega} D_{r_1 r_2} - M \sum_{k \in \mathcal{M}} u_{ijk}, \qquad \forall i, j \in N,\ i \neq j,\ \forall \omega \in \Omega \in \Omega, \tag{A.14}$$

$$x_{ik} - x_{jk} \leq d_{ij}^{\text{int},x} + \bar{L} \cdot \left(1 - \sum_{t \in \mathcal{M}} u_{ijt}\right) \qquad \forall i, j \in N, i \neq j,\ \forall k \in \mathcal{M}, \tag{A.15}$$

$$x_{jk} - x_{ik} \le d_{ij}^{\text{int},x} + \bar{L} \cdot (1 - \sum_{t \in \mathcal{M}} u_{ijt}) \quad \forall i,j \in N, i \ne j, \ \forall k \in \mathcal{M}, \tag{A.16}$$

$$y_{ik} - y_{jk} \le d_{ij}^{\text{int},y} + \bar{W} \cdot \left(1 - \sum_{t \in \mathcal{M}} u_{ijt}\right) \quad \forall i,j \in N, i \ne j, \ \forall k \in \mathcal{M}, \tag{A.17}$$

$$y_{jk} - y_{ik} \le d_{ij}^{\text{int},y} + \bar{W} \cdot \left(1 - \sum_{t \in \mathcal{M}} u_{ijt}\right) \quad \forall i,j \in N, i \ne j, \ \forall k \in \mathcal{M}, \tag{A.18}$$

$$2\bar{L}\sum_{t \in \mathcal{M}} u_{ijt} + d_{ij}^{\text{int},x} \ge x_{ik} + x_{jg} \quad \forall i,j \in N, i \ne j, \ \forall k \ne g \in \mathcal{M}, \tag{A.19}$$

$$2\bar{W}\sum_{t \in \mathcal{M}} u_{ijt} + d_{ij}^{\text{int},y} \ge y_{ik} + y_{jg} \quad \forall i,j \in N, i \ne j, \ \forall k \ne g \in \mathcal{M}, \tag{A.20}$$

$$x_{ik} + 0.5 l_i Y_{ik} \le x_{jk} - 0.5 l_j Y_{jk} + \bar{L}(2 - \alpha_{ijk} - \beta_{ijk}) \quad \forall i,j \in N, i \ne j, \ \forall k \in \mathcal{M}, \tag{A.21}$$

$$y_{ik} + 0.5 w_i Y_{ik} \le y_{jk} - 0.5 w_j Y_{jk} + \bar{W}(1 + \alpha_{ijk} - \beta_{ijk}) \quad \forall i,j \in N, i \ne j, \ \forall k \in \mathcal{M}, \tag{A.22}$$

$$\sum_{k \in \mathcal{M}} Y_{ik} = 1 \quad \forall i \in N, \tag{A.23}$$

$$T_k \le T_{k-1} \quad \forall k \in \mathcal{M}, k > 1, \tag{A.24}$$

$$\sum_{i \in N} Y_{ik} \ge T_k \quad \forall k \in \mathcal{M}, \tag{A.25}$$

$$T_k \cdot |N| \ge \sum_{i \in N} Y_{ik} \quad \forall k \in \mathcal{M}, \tag{A.26}$$

$$u_{ijk} \le 0.5(Y_{ik} + Y_{jk}) \quad \forall i,j \in N, i \ne j, \ \forall k \in \mathcal{M}, \tag{A.27}$$

$$u_{ijk} + 1 \ge Y_{ik} + Y_{jk} \quad \forall i,j \in N, i \ne j, \ \forall k \in \mathcal{M}, \tag{A.28}$$

$$0.5 l_i Y_{ik} \le x_{ik} \le (\bar{L} - 0.5 l_i) Y_{ik} \quad \forall i \in N, \ \forall k \in \mathcal{M}, \tag{A.29}$$

$$0.5 w_i Y_{ik} \le y_{ik} \le (\bar{W} - 0.5 w_i) Y_{ik} \quad \forall i \in N, \ \forall k \in \mathcal{M}, \tag{A.30}$$

$$\alpha_{ijk} + \alpha_{jik} = 1, \quad \beta_{ijk} + \beta_{jik} = 1 \quad \forall i,j \in N, i \ne j, \ \forall k \in \mathcal{M}, \tag{A.31}$$

$$\alpha_{isk} + \alpha_{sjk} - \alpha_{ijk} \le 1, \quad \beta_{isk} + \beta_{sjk} - \beta_{ijk} \le 1 \quad \forall i,s,j \in N \text{ and distinct}, \ \forall k \in \mathcal{M}, \tag{A.32}$$

$$Y_{ik},\ T_k,\ u_{ijk},\ Z^{\omega}_{ijk_1k_2r_1r_2},\ \alpha_{ijk},\ \beta_{ijk} \in \{0,1\}, \tag{A.33}$$

$$x_{ik},\ y_{ik},\ d_{ij}^{\text{int},x},\ d_{ij}^{\text{int},y},\ d_{ij}^{\text{ext},\omega},\ v_{kr}^{\omega},\ F^{\omega}_{ijk_1k_2r_1r_2},\ C^{\omega},\ \delta^{+,\omega},\ \delta^{-,\omega},\ \bar{C},\ V \ge 0. \tag{A.34}$$

Objective Function (A.1) minimizes the sum of expected transportation cost, cost deviation penalty, and ramp violation penalty. Equation (A.2) defines the expected cost as the probability-weighted sum of scenario costs. Equation (A.3) captures positive and negative deviations of each scenario cost from the expected cost. Equation (A.4) aggregates ramp capacity violations across all scenarios, blocks, and ramps. Equation (A.5) decomposes each scenario cost into internal transport, external transport, and closeness penalty components, defined in (A.6)–(A.8). Constraint (A.9) ensures all interblock flow between departments i and j is fully routed through available ramp pairs. Constraint (A.10) restricts flow to routes where i and j are assigned to blocks k_1 and k_2, respectively. Constraint (A.11) limits outbound flow through each ramp to its scenario-specific capacity, with excess captured by v_{kr}^{ω}. Constraint (A.12) links the flow variable to the binary route indicator via big-M. Constraint (A.13) forces external distance to zero when departments share a block. Constraint (A.14) computes weighted external distance based on ramp-to-ramp distances. Constraints (A.15)–(A.18) compute Manhattan distance components between departments within the same block. Constraints (A.19)–(A.20) bound internal distances when departments are in different blocks. Constraints (A.21)–(A.22) enforce nonoverlap between departments within the same block. Constraint (A.23) assigns each department to exactly one block. Constraints (A.24)–(A.26) enforce sequential block usage. Constraints (A.27)–(A.28) link the colocation variable u_{ijk} to block assignments. Constraints (A.29)–(A.30) ensure departments fit within block boundaries. Constraints (A.31)–(A.32) enforce symmetry and transitivity of directional positioning variables. Constraints (A.33)–(A.34) define variable domains.

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Verification Letter

Armağan Aktop, Director, Supplier Development & Digital Transformation, Beko, Beyoğlu 34445 Istanbul, Turkey, writes: "This letter is to confirm that the study presented in the manuscript titled "Designing Resilient Factory Layouts Through Data-Driven Scenario Analysis with an Industrial Application," authored by Sadra Shoarinejad et al., was developed in collaboration with Beko (Arçelik A.Ş.), specifically considering one of our washing machine production plants.

"The proposed approach addresses a facility layout planning problem that reflects our operational environment. The results obtained from the study have provided valuable insights for the redesign and reconfiguration of the facility layout.

"We support the submission of this work to the *INFORMS Journal on Applied Analytics*."

Sadra Shoarinejad is a researcher at the Supplier Development and Digital Transformation Team at Beko Corporate. He works on the development of optimization and simulation-based methods for performance improvement of suppliers and supplier facility layout design. He holds a PhD degree in industrial engineering from Koç University.

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